ADDITIONAL PRACTICE PROBLEMS

1. A shopkeeper marks an item 40% above the cost price and allows a discount of 25%. What is his profit percentage?

2. In a 60-liter mixture of milk and water, the ratio of milk to water is 2:1. How much water must be added to make the ratio 1:2?

3. A trader buys 80 pens for ₹400 and sells them at ₹6 each. Find his profit or loss percentage.

4. A shopkeeper earns 25% profit on selling an article at 20% discount. Find the markup % on cost price.

5. A man bought an article for ₹240 and sold it at a loss of 12.5%. What is the selling price?

6. 20 L of a mixture contains milk and water in the ratio 4:1. How much water must be added to make the ratio 4:3?

7. A container has milk and water in the ratio 7:5. 6L of the mixture is replaced by water. The new ratio is?

8. A mixture contains juice and water in the ratio 3:2. If 10 liters of water is added, the ratio becomes 3:4. Find the original quantity.

9. An article is sold at 25% gain. If the selling price is ₹500, find the cost price.

10. A person sold a watch at a gain of 12%. If the cost price was ₹1500, what was the selling price?

11. A trader mixes tea costing ₹60/kg with tea costing ₹80/kg in ratio 3:2 and sells the mixture at ₹78/kg. Find profit %.

12. 40 kg rice at ₹30/kg is mixed with 60 kg rice at ₹50/kg. The mixture is sold at ₹48/kg. Find profit %.

13. If a shopkeeper makes a loss of 5% on selling an item for ₹380, what is the cost price?

14. If an article is sold at a loss of 12% and the selling price is ₹440, find the cost price.

15. If a shopkeeper makes a loss of 5% on selling an item for ₹380, what is the cost price?

16. Two trains of lengths 150 m and 100 m are moving in opposite directions at 60 km/h and 90 km/h. How long will they take to cross each other?

17. If a man walks at 5 km/hr, he misses the train by 10 minutes. If he walks at 6 km/hr, he reaches 5 minutes early. Find the distance to the station.

18. A man rows 15 km upstream in 3 hours and downstream in 1.5 hours. Find the speed of the boat and the stream.

19. A cyclist travels from A to B at 20 km/h and returns at 25 km/h. The total time taken is 9 hours. Find the distance between A and B.

20. A bike takes 2 hours to cover a certain distance. If its speed is increased by 10 km/hr, it takes 1.5 hours. Find the original speed.

21. Solve: 3x+4y<123x + 4y < 123x+4y<12 and x+y>2x + y > 2x+y>2

22. If x+y=10x + y = 10x+y=10, and x−y>4x - y > 4x−y>4, what are the possible values of x?

23. A man wants to buy x pens and y notebooks. Each pen costs ₹5 and each notebook ₹8. He has ₹100. Write the inequality and find feasible pairs.

24. Divide ₹7600 among A, B, and C in the ratio 4:5:6.

25. In a mixture of 60 liters, the ratio of acid to water is 3:2. 10 liters of water is added. What is the new ratio?

26. A, B, and C share profit in the ratio 5:3:2. If the total profit is ₹8600, how much does B get?

27. Solve: 2x−3 /5 = x+1 /2

28. Solve the quadratic: x^2 – 7x + 12 = 0

29. Factorise: x^2 - 2x – 15

30. Factorise: x^2 - 10x + 25

31. Simplify: (x+3)^2−16

32. A and B run around a circular track of 300 m in 24s and 18s. After how many seconds will they meet at the starting point?

33. A runs a circular track in 30s, B in 36s, C in 45s. After how many seconds will they meet together at the start?

34. A and B start from the same point and move in opposite directions around a circle of 300 m. Their speeds are 6 m/s and 9 m/s. After how much time will they meet?

35. If the 5th term of an AP is 22 and 15th term is 62, find the first term and common difference.

36. Which term of the AP 3, 7, 11, ... is 95?

37. The sum of first n terms of an AP is 3n2+2n3n^2 + 2n3n2+2n. Find the 10th term.

38. A milkman adds 2 liters of water to 18 liters of milk and sells the entire mix at milk's cost price. What is his gain %?

39. A container has 40 L of a 3:1 acid-water mixture. How much water should be added to make the ratio 2:3?

40. Find the greatest number that will divide 134, 298, and 462 leaving the same remainder.

41. Find the largest number which exactly divides 1251, 9377, and 15628.

42. The sum of the 5th and 9th terms of an AP is 72. If the 15th term is 2 more than twice the 7th term, find the first term and the common difference.

43. If the 3rd, 6th, and 9th terms of an AP are in the ratio 1:2:3, find the ratio of the first term to the common difference.

44. The 8th term of an AP is 5 times the 2nd term, and the 11th term exceeds the 5th term by 36. Find the first term and common difference

45. A person saves ₹500 in the first month, and each month he increases his saving by ₹100. How much does he save in a year?

46. A factory produces two products A and B. Each unit of A requires 3 hours of labor and 2 kg of material. Each unit of B requires 4 hours of labor and 3 kg of material. The factory has at most 60 labor hours and 40 kg of material available. Form the inequalities and find the feasible region for number of units of A and B.

47. A company plans to hire junior and senior developers. A junior developer works 5 hours/day and earns ₹200/day. A senior works 8 hours/day and earns ₹500/day. The company has a maximum of 80 working hours and a budget of ₹4000 per day. Form inequalities

48. A diet plan must include at least 40g protein and 60g carbs. Food X gives 10g protein and 20g carbs per unit; Food Y gives 20g protein and 10g carbs per unit. A person can consume at most 6 units in total. Find all combinations of X and Y that meet the requirements.

49. Find the greatest number that will divide 134, 298, and 462 leaving the same remainder.

50. Simplify:

$$\frac{x^2-16}{x+4}$$

51. Factor and simplify:

$$\frac{x^2+5x+6}{x^2-9}$$

52. Simplify:

$$\frac{x^2-1}{x^2+2x+1}$$

53. Reduce:

$$\frac{x^2+x-6}{x^2-2x-3}$$